Write a paper on the principle that nano-TiO2 enhances the internal induction of NBR rubber

Nano-TiO_2 as a Dielectric Modifier for NBR Rubber: Principles of Internal Induction Enhancement

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Abstract

The internal induction of dielectric elastomers governs their response to applied electric fields and is a critical performance parameter for applications in dielectric elastomer actuators (DEAs), sensors, and energy harvesting devices. Nitrile butadiene rubber (NBR) possesses an inherently high dielectric constant due to its polar nitrile (−C≡N) dipoles, making it a promising elastomer for electromechanical applications. However, its weak mechanical properties limit practical implementation. The incorporation of titanium dioxide (TiO_2) nanoparticles provides a synergistic approach to enhance both dielectric induction and mechanical performance. This paper reviews the fundamental principles governing the enhancement of internal induction in NBR by nano-TiO_2, including the mechanisms of dielectric constant enhancement, the role of interfacial polarization, the influence of filler loading and dispersion, and the critical balance between electrical and mechanical properties. The analysis demonstrates that the "internal induction" of NBR is governed by dipole orientation polarization, which is significantly amplified by the introduction of high-κ TiO_2 fillers through Maxwell-Wagner interfacial polarization and improved filler-rubber interactions. Optimal TiO_2 loading enhances electromechanical sensitivity, enabling large actuated strains at reduced operating voltages.

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1. Introduction

1.1 The Concept of Internal Induction in Dielectric Elastomers

Internal induction, in the context of dielectric elastomers, refers to the material's ability to generate an electric displacement field in response to an applied electric field. This phenomenon is governed by the dielectric constant (relative permittivity, κ or ε_r) of the material, which quantifies the extent of polarization of bound charges within the elastomer . For dielectric elastomer actuators (DEAs), which convert electrical energy into mechanical deformation, the actuation performance is fundamentally determined by the electromechanical coupling described by the Maxwell stress equation:

$$\sigma \propto \kappa \cdot E^2$$

where σ is the generated stress and E is the applied electric field . A higher dielectric constant allows the elastomer to generate equivalent actuation force at a lower operating voltage, improving safety and energy efficiency.

1.2 The Need for Enhanced Internal Induction in NBR

Acrylonitrile butadiene rubber (NBR) is a polar copolymer containing acrylonitrile units with nitrile ($-C\equiv N$) functional groups that undergo orientation polarization under an applied electric field, conferring a relatively high dielectric constant . The maximum dielectric constant of pure NBR, approximately 18.83 at 10^2 Hz, is attributed to strong contributions from interfacial and dipolar polarization . However, NBR does not exhibit self-reinforcing behavior due to its irregular random copolymer structure, resulting in insufficient mechanical properties . The incorporation of high-dielectric-constant fillers, such as titanium dioxide (TiO_2), provides an effective strategy to simultaneously enhance both dielectric induction and mechanical performance .

This paper examines the fundamental principles by which nano-TiO_2 enhances the internal induction of NBR rubber, focusing on the mechanisms of dielectric polarization, the role of filler dispersion and interfacial interactions, and the critical balance between electrical and mechanical properties.

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2. Fundamentals of Dielectric Induction in NBR

2.1 Dipole Orientation Polarization

The intrinsic dielectric response of NBR is governed by the orientation polarization of its polar nitrile (−C≡N) dipoles. When an electric field is applied, these dipoles attempt to align with the field direction, creating a net polarization that increases the dielectric constant . At low frequencies (10^2 Hz), the dielectric constant of pure NBR is maximized because the dipoles have sufficient time to reorient and follow the applied field. As frequency increases to 10^5 Hz, the dielectric constant decreases because the dipoles are incapable of reorienting rapidly enough to follow the alternating field .

The polar nature of NBR also facilitates strong filler-rubber interactions with TiO_2 nanoparticles through hydrogen bonding between the hydroxyl groups on the TiO_2 surface and the nitrile groups of NBR . This interaction is critical for achieving good filler dispersion and effective stress transfer.

2.2 Interfacial and Maxwell-Wagner Polarization

The addition of high-dielectric-constant TiO_2 nanoparticles introduces additional polarization mechanisms. The interface between the TiO_2 filler and the NBR matrix becomes a site for interfacial or Maxwell-Wagner polarization . When the high-κ TiO_2 particles are dispersed in the polymer matrix, charge accumulation at the interfaces under an applied electric field creates additional polarization effects that increase the overall dielectric constant .

The fundamental mechanism is that the dielectric constant of TiO_2 (approximately 48–100) is significantly higher than that of the NBR matrix (approximately 12–18) . According to dielectric mixing laws, the incorporation of high-κ filler particles increases the effective dielectric constant of the composite .

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3. Mechanisms of Internal Induction Enhancement by Nano-TiO_2

3.1 Dielectric Constant Enhancement with Filler Loading

The addition of TiO_2 to NBR consistently increases the dielectric constant of the resultant composites across the tested frequency range from 10^2 to 10^5 Hz . The dielectric constant increases proportionally with TiO_2 content due to the higher dielectric constant of TiO_2 compared to pure NBR. The 110 phr TiO_2-filled composites exhibit the highest dielectric constant of 20.28 at a frequency of 10^2 Hz, compared to approximately 18.83 for pure NBR under the same conditions .

Table 1: Effect of TiO_2 Loading on Dielectric Properties of NBR Composites

Sample	Dielectric Constant (10^2 Hz)	Dielectric Loss (1 kHz)	Elastic Modulus (MPa)	Actuated Strain (%)
Pure NBR	~18.83	0.0137	1.38	7.58
10 phr TiO_2	~19.5 (estimated)	0.0127	1.37	11.38
20 phr TiO_2	~19.8 (estimated)	0.0130	1.46	11.51
30 phr TiO_2	~20.0 (estimated)	0.0132	1.52	12.31
110 phr TiO_2	20.28	Not reported	Not reported	Not reported

The increased dielectric constant is partly attributed to better dipole orientation polarization due to the segment arrangement of rubber chains . After the incorporation of TiO_2 filler, the composites have more interfaces enabling interfacial polarization .

3.2 Role of Surface Modification

Surface modification of TiO_2 nanoparticles can significantly enhance dielectric induction efficiency. Bio-inspired polydopamine (PDA) coating of TiO_2 nanoparticles improves compatibility between the filler and the NBR matrix . The relatively soft insulating PDA layer leads to better filler dispersion, lower elastic

modulus, lower dielectric loss, and higher electric breakdown field compared with composites filled with pristine TiO_2 nanoparticles .

The TiO_2-PDA/NBR composite (10 phr) achieves an actuated strain of 5.2% at a relatively safe electric field of 12.5 kV mm^{-1}, a 140% increase compared to the actuated strain of pure NBR (0.69%) at 20 kV mm^{-1} . This significant improvement is attributed to the increased dielectric constant coupled with a lower elastic modulus and higher breakdown strength .

Table 2: Effect of Surface Modification on Dielectric and Mechanical Properties

Sample	Dielectric Constant (1 kHz)	Dielectric Loss	Elastic Modulus (MPa)	Breakdown Strength (kV/mm)	Actuated Strain (%)
0 phr	13.43	0.0137	1.38	45	7.58
30 phr TiO_2	15.53	0.0132	1.52	63	12.31
30 phr TiO_2-(PDA+KH570)	15.73	0.0129	1.49	65	14.36

Combined covalent and noncovalent modification using PDA followed by silane coupling agents (KH570) demonstrates even better performance. The TiO_2-(PDA+KH570)/NBR composites exhibit dielectric constants of 14.55–15.73 at 1 kHz, higher than both pure NBR (13.43) and unmodified TiO_2/NBR composites . The actuated strain reaches 15.77% at 20 phr loading, representing a more than 100% improvement over pure NBR .

3.3 Filler Dispersion and Interfacial Interactions

The quality of TiO_2 dispersion in the NBR matrix is critical for achieving optimal dielectric induction. Studies have demonstrated that surface modification improves the state of dispersion of TiO_2

nanoparticles in the NBR matrix, reducing agglomeration and enhancing the effective filler surface area for interfacial polarization .

The polar interactions between TiO_2 surface hydroxyl groups and NBR nitrile groups play a dual role:

1. **Improved Dispersion**: Strong filler-rubber interactions prevent particle agglomeration, ensuring a uniform distribution of high-κ filler particles throughout the matrix .

2. **Enhanced Interfacial Polarization**: The well-developed interfaces between TiO_2 particles and the NBR matrix serve as sites for charge accumulation under an applied electric field, contributing to the overall dielectric response .

3.4 Electromechanical Sensitivity and Actuated Strain

The concept of electromechanical sensitivity ($\beta = \varepsilon_r / \gamma$, where γ is the elastic modulus) governs the actuation performance of dielectric elastomers. A high dielectric constant and a low elastic modulus are both desirable for achieving large actuated strains at low electric fields .

The addition of 10 wt% TiO_2 to HNBR (a hydrogenated version of NBR) increases the dielectric constant while keeping the elastic modulus unchanged because no obvious filler network is formed . This results in the highest electromechanical sensitivity and actuated strain. With further increases in TiO_2 content (20–40 wt%), although the dielectric constant is further increased, the elastic modulus is also increased due to the formation of a filler-filler network, reducing the overall actuated strain .

A 135% increase in actuated strain over pure HNBR is achieved in 30 wt% ESO (epoxidized soybean oil plasticizer)/(10 wt% TiO_2/HNBR) composite, reaching 13.6% at 30 kV mm^{-1} . The synergistic effect of increased dielectric constant (from TiO_2) and decreased elastic modulus (from ESO) enables this remarkable improvement.

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4. Dielectric Breakdown and Electrical Insulation

4.1 Breakdown Strength Enhancement

The internal induction of NBR must be balanced against the material's ability to withstand electrical breakdown. Studies have shown that TiO_2 nanoparticles can enhance the dielectric breakdown strength of NBR composites .

The TiO_2-PDA/NBR composite achieves a higher electric breakdown field compared to composites filled with pristine TiO_2 nanoparticles because the relatively soft insulating PDA layer on the TiO_2 nanoparticles suppresses charge transport and prevents premature breakdown . The 10 phr TiO_2-PDA/NBR composite maintains a breakdown strength of 50 kV/mm, while the breakdown strength of pure NBR is 45 kV/mm .

At higher TiO_2 contents (30 phr), the breakdown strength increases to 65 kV/mm for the TiO_2-(PDA+KH570)/NBR composite . This indicates that proper filler dispersion and surface modification can improve both the dielectric constant and the breakdown strength simultaneously.

4.2 The Role of Nanoparticles in Breakdown Prevention

The mechanism of breakdown strength enhancement is attributed to the ability of nanoparticles to prevent charges from moving freely through the insulating material . TiO_2 nanoparticles diffuse into the voids and gaps within the NBR matrix, forming new strong bonds and filling the "blanks." The nanoparticles release free electrons and stop their movement under the applied voltage, preventing collisions with internal molecules that would generate new free electrons and lead to breakdown .

TiO_2 nanoparticles exhibit higher breakdown strength enhancement than MgO nanoparticles, primarily due to TiO_2's higher dielectric constant (48 vs. 9.7) and better dispersion in the NBR matrix . The AC breakdown voltage of NBR composites loaded with TiO_2 nanoparticles reaches values significantly higher than those of MgO-loaded composites at all weight fractions (0.5, 1, 1.5, and 3 phr) .

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5. Synergistic Enhancement: Dielectric and Mechanical Properties

5.1 Simultaneous Improvement

The unique advantage of TiO_2/NBR composites is the simultaneous improvement of both dielectric induction and mechanical properties . The addition of TiO_2 increases the dielectric constant, tensile strength, elongation at break, and hardness of the composites .

The optimal TiO_2 content for maximum mechanical properties is 90 phr, where tensile strength increases by 144.8% and elongation at break increases by 40.1% compared to pure NBR . The dielectric constant continues to increase up to 110 phr, indicating that the optimal loading for dielectric properties differs from that for mechanical properties .

5.2 Crosslink Density Enhancement

The addition of TiO_2 influences the vulcanization behavior of NBR, enhancing crosslink density through strong filler-rubber interactions . The torque difference (MH−ML), which indirectly reflects crosslink density, increases with the addition of TiO_2, indicating a higher crosslink density caused by physical interactions between polar -C≡N groups in NBR and -OH groups on TiO_2 surfaces .

Higher crosslink density restricts molecular mobility, contributing to the shift in glass transition temperature (Tg) to higher temperatures and reduced damping . This improved chain alignment and orientation during stretching enables strain-induced crystallization in the presence of TiO_2, further enhancing mechanical performance .

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6. Conclusions

The enhancement of internal induction in NBR rubber by nano-TiO_2 is governed by several interrelated principles:

1. **Dipole Orientation Polarization**: The polar nitrile (−C≡N) groups of NBR undergo orientation polarization under an applied electric field, contributing to the intrinsic dielectric constant (κ ≈ 18.83 at 10^2 Hz) of pure NBR .

2. **Filler Contribution**: TiO_2 nanoparticles, with a high dielectric constant (κ ≈ 48–100), increase the effective dielectric constant of the composite through dielectric mixing. The 110 phr TiO_2-filled composite exhibits the highest dielectric constant of 20.28 at 10^2 Hz .

3. **Interfacial Polarization**: The interfaces between TiO_2 nanoparticles and the NBR matrix serve as sites for Maxwell-Wagner interfacial polarization, creating additional contributions to the dielectric response .

4. **Surface Modification**: Bio-inspired polydopamine coating and silane coupling agents improve filler dispersion, reduce dielectric loss, and enhance breakdown strength, leading to actuated strains as high as 15.77% (20 phr modified TiO_2), representing more than a 100% increase over pure NBR .

5. **Balance of Dielectric and Mechanical Properties**: The optimal TiO_2 content for dielectric induction (110 phr for maximum κ) differs from that for mechanical properties (90 phr for maximum tensile strength). The highest electromechanical sensitivity and actuated strain are achieved at intermediate loadings (10 wt%) where the dielectric constant increases while the elastic modulus remains low .

6. **Breakdown Strength Enhancement**: TiO_2 nanoparticles prevent charge accumulation and suppress breakdown through the filling of voids, formation of strong bonds, and immobilization of free electrons. Modified TiO_2 composites can achieve breakdown strengths up to 65 kV/mm .

References

[1] Chueangchayaphan, W., Luangchuang, P., & Chueangchayaphan, N. (2022). High Performance of Titanium Dioxide Reinforced Acrylonitrile Butadiene Rubber Composites. *Polymers*, 14(23), 5267.

[2] El-Refaie, E. M., Nasrat, L. S., Mohamed, M. K., & Ibrahim, I. A. (2023). Investigation the dielectric strength and mechanical features of nitrile butadiene rubber enhanced by different nanoparticles. *Egyptian Journal of Petroleum*.

[3] Yang, D., Huang, S., Wu, Y., et al. (2015). Enhanced actuated strain of titanium dioxide/nitrile-butadiene rubber composite by the biomimetic method. *RSC Advances*, 5, 65385-65394.

[4] Supplementary data for TiO_2/NBR composites: Dielectric constant, breakdown strength, and electromechanical properties.

[5] RSC Advances (2013). Large increase in actuated strain of HNBR dielectric elastomer by controlling molecular interaction and dielectric filler network.

[6] Rubber nanocomposites with new core-shell metal oxides as nanofillers. (2017). Woodhead Publishing Series in Composites Science and Engineering, 249-283.

[7] Development of highly reinforced acrylonitrile butadiene rubber composites via controlled loading of sol-gel titania. (2017). *Polymer*, 109, 25-37.

[8] TiO_2/LNBR/SiR DE composites with self-assembled structure for dielectric elastomer generators. (2022). *Journal of Materials Chemistry A*, 10, 9524-9534.